DATA CENTER SOLUTIONS

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INSIGHT · PLANNING & POWER

DATA CENTER SITE SELECTION AND POWER PLANNING: SETTLE THE DECISION BASIS BEFORE YOU SIGN ANYTHING

Power may be the first major constraint in a data-center site decision, but it is not the only one. A credible site evaluation connects electrical availability and timing with land, permitting, water, fiber, construction access, community considerations, phasing and the operating requirement.

JLL's 2026 Global Data Center Outlook describes power as the primary site-selection criterion. CBRE's North America Data Center Trends report also emphasizes power availability and infrastructure delivery. These are market conditions, not evidence that every site with a power path is viable.

START WITH THE FACILITY REQUIREMENT

Before comparing sites, define what the program is trying to place into operation.

- What workload and initial capacity are being planned?
- How may capacity change by phase?
- What availability and resilience requirements apply?
- When must decisions—not just operations—occur?
- Which assumptions are approved, and which remain provisional?

Without that basis, teams can compare sites against moving criteria.

SEPARATE POWER AVAILABILITY FROM A COMPLETE POWER PLAN

A statement that power is “available” does not answer every project question. The team should document the source, amount, timing, conditions, infrastructure work, approvals, interfaces and owner of each required action.

The power plan also needs to connect to facility design, equipment, construction, commissioning and maintenance. A decision that works on a planning spreadsheet may still create physical, operational or schedule dependencies elsewhere.

EVALUATE THE WHOLE SITE SYSTEM

The site screen should identify which subjects can stop or materially change the project:

1. Power availability, timing and infrastructure dependencies.
2. Land, access, grading and construction logistics.
3. Permitting, entitlement and jurisdictional requirements.
4. Water and thermal-system implications.
5. Fiber and network requirements.
6. Environmental and community considerations, including noise and visible infrastructure.
7. Capacity phasing and future expansion.
8. Commissioning, operations and long-term serviceability.

The responsible discipline and evidence required for each decision should be explicit. Do not assume one provider owns every item.

USE DECISION GATES—NOT A SINGLE FINAL SCORE

A weighted score can help compare options, but it should not hide a fatal constraint. Establish gates for evidence that must exist before the program advances: confirmed requirements, credible power assumptions, jurisdictional review, site-control terms, infrastructure responsibilities and an approved next-phase basis.

Record who made each decision, what evidence was used and what remains open. That record becomes part of the handoff into design and preconstruction.

WHERE EVOLVE FITS

Evolve supports planning as part of the Plan, Design, Build, Power and Maintain lifecycle. its documented capability also includes design, procurement, construction, commissioning, power conversions, monitoring and maintenance. The exact site-selection, utility, permitting or infrastructure responsibilities must be defined for the engagement.

EXPLORE PLANNING & FEASIBILITY

EXPLORE POWER GENERATION

DIRECT ANSWERS

WHAT IS THE FIRST QUESTION IN DATA-CENTER SITE SELECTION?

Define the facility and capacity requirement, then determine whether a credible power path and the wider site conditions can support it.

IS POWER AVAILABILITY ENOUGH TO QUALIFY A SITE?

No. Timing, conditions, infrastructure work, approvals, land, water, fiber, logistics, community factors, phasing and operating needs also affect viability.

WHAT SHOULD A SITE-SELECTION DECISION RECORD CONTAIN?

It should identify the requirement, evidence, assumptions, constraints, responsibilities, open questions and the decision gate for moving forward.

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SOURCES

- JLL, 2026 Global Data Center Outlook (January 2026)
- CBRE, North America Data Center Trends H1 2026 (August 27, 2026)